# Physics reference

## SI units

All scientists use seven basic units of measurement, known as the SI base units, listed below. "SI" stands for "Système International." The units are maintained by experts in the headquarters, located in Paris, France.

**SI UNITS**

| Unit | Symbol | Quantity measured |
|---|---|---|
| meter | m | unit of length, defined as the distance light travels through a vacuum in 1/299,792,458th of a second |
| kilogram | kg | unit of mass, defined by the International Standard Kilogram made of a platinum-iridium alloy in Paris, France |
| second | s | unit of time, defined in terms of the frequency of a type of light radiated by a cesium atom |
| ampere | A | unit of electrical current, defined by the attraction force between two parallel conductors that are conducting one ampere |
| kelvin | K | unit on a scale of temperature that begins at absolute zero: 0 Kelvin or 459.67˚F (-273.15˚C) |
| candela | cd | a measure of luminous intensity (how powerful a light source is); one candle has a luminous intensity of one candela |
| mole | mol | a unit of quantity of a substance (generally very small particles such as atoms and molecules); one mole is made up of 6.02 x $10^{23}$ objects (atoms or molecules) |

## Derived SI units

This table contains just a few units that are derived from combinations of the seven base SI units. Nevertheless these units are very widely used and have been given their own names.

**SI UNITS**

| Unit | Symbol | Quantity measured |
|---|---|---|
| becquerel | Bq | unit of radioactive decay; the quantity of material in which one nucleus decays per second |
| Celsius | ˚C | unit of temperature, with the same magnitude as a Kelvin, but zero is at water's freezing point |
| coulomb | C | closely related to an ampere, this is the quantity of charge carried each second by a current of one ampere |
| farad | F | unit of capacitance, which is a capacitor's ability to store charge |
| hertz | Hz | unit of frequency; the number of cycles or repeating events per second |
| joule | J | amount of energy transferred when a force of one newton is applied over one meter |
| newton | N | unit of force required to increase the velocity of a mass by 1 kg by 1 m per second every second |
| ohm | Ω | unit of resistance; a one ohm resistor allows a current of one ampere to flow when one volt is applied across it |
| pascal | Pa | unit of pressure; a pascal is a force of one newton applied across an area of one square meter |
| volt | V | unit of potential difference and the force that pushes electric current |
| watt | W | unit of power (the rate at which energy is expended); calculated as joules per second |